Yarone Meir Tokayer

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EDUCATION

Yale University — New Haven, CT
Ph.D., Physics — Expected 2027
Advisor: Frank van den Bosch.
Thesis: *Probing collisional gravitational systems with the gravothermal fluid model*.
Certificate in Public Communication — Apr. 2026
M.S. and M.Phil. — May 2025

Columbia University — New York, NY
M.A., Philosophical Foundations of Physics — Feb. 2020
Advisor: David Z. Albert.
Thesis: *Probability in Everettian Quantum Mechanics*.

The Cooper Union — New York, NY
B.S., Engineering, *summa cum laude*; Minor in Mathematics — May 2014
Senior project: *Muscle Denervation and Neurotensin as a Therapeutic Target for ALS*.

RESEARCH INTERESTS

Theoretical and computational astrophysics; gravitational dynamics; stellar systems; self-interacting dark matter; collisional dynamics; numerical modeling.

PUBLICATIONS

A complete list of publications can also be found on Google Scholar.

Refereed Journal Publications

1. **Tokayer, Y. M.**, Koss, M., Urry, C. M., et al., *ApJ*, 982, 134 (2025). doi:10.3847/1538-4357/adb8c9.
2. **Tokayer, Y. M.**, Dutra, I., Natarajan, P., et al., *ApJ*, 970, 143 (2024). doi:10.3847/1538-4357/ad51fd.
3. Guolo, M., Gezari, S., Yao, Y., et al. (incl. **Tokayer, Y. M.**), *ApJ*, 966, 160 (2024). doi:10.3847/1538-4357/ad2f9f.
4. Burgess, D., Mori, K., Gelfand, J. D., et al. (incl. **Tokayer, Y. M.**), *ApJ*, 930, 148 (2022). doi:10.3847/1538-4357/ac650a.
5. **Tokayer, Y. M.**, An, H., Halpern, J. P., et al., *ApJ*, 923, 17 (2021). doi:10.3847/1538-4357/ac2c6a.

Submitted and In Preparation

1. van den Bosch, F. C. & **Tokayer, Y. M.**, "Not so fast: revised calibration of SIDM core collapse." *Accepted for publication at MNRAS.*
2. **Tokayer, Y. M.**, van den Bosch, F. C., & Errani, R., "Sink or swim: a fast gravothermal fluid model for multi-component stellar systems." *In preparation.*
3. **Tokayer, Y. M.** & van den Bosch, F. C., "The gravothermal collapse of SIDM halos is not a viable process for producing massive black hole seeds." *In preparation.*
4. van den Bosch, F. C., & **Tokayer, Y. M.**, "Through ups and downs: transient lensing magnification during SIDM core collapse." *In preparation.*

AWARDS AND SCHOLARSHIPS

- **Finalist, 3-Minute Thesis Competition**, Yale Graduate School of Arts and Sciences, 2026.
- **Leigh Page Award for Excellence in Graduate Student Teaching**, Yale Physics Department, 2025.
- **Teacher Award**, Robotraffic Competition, Technion, Haifa, Israel, 2017; coach of the SAR High School team.
- **Entrance Scholarship**, Philosophical Foundations of Physics Program, Columbia University, Fall 2016.
- **Goodman Prize for Essay in the Humanities**, The Cooper Union, Spring 2013.
- **Tau Beta Pi**, Engineering Honor Society.
- **Dean's List**, The Cooper Union, all semesters.

RESEARCH EXPERIENCE

Graduate Research Assistant, Yale University

Gravothermal evolution of dark matter halos and stellar systems Jun. 2024–Present
PI: Frank van den Bosch

- Develop gravothermal fluid models of multicomponent stellar systems and self-interacting dark matter (SIDM) halos to study core collapse and mass segregation.
- Design simulation suites to test SIDM black hole seeding scenarios and investigate the dynamical evolution of collisional gravitational systems.
- Implement idealized N-body simulations of cold dark matter halo growth and compare their evolution with cosmological predictions.

Selection effects in deep-field AGN surveys Jul. 2021–Aug. 2024
PIs: Michael Koss and C. Megan Urry

- Developed a simulation pipeline using BASS to generate and fit high-redshift *Chandra* active galactic nucleus (AGN) spectra.
- Analyzed 2,800 simulated spectra to quantify observational selection biases; published results in *ApJ* and presented a poster at the 241st AAS meeting.

Dark matter profiles from cluster lensing Jan. 2022–May 2024
PI: Priyamvada Natarajan

- Developed algorithms to fit cluster dark matter subhalo profiles to galaxy–galaxy strong-lensing measurements.
- Derived an empirical concentration–mass relation to assess tensions with ΛCDM predictions; published results in *ApJ*.

Research Assistant, Columbia University

Spectral, imaging, and timing analysis of X-ray observations Mar. 2020–Dec. 2020
PIs: Charles Hailey and Kaya Mori

- Analyzed *NICER* observations of AR Scorpii, measuring its spin period with sub-microsecond precision and identifying pulsed nonthermal X-ray emission.
- Analyzed *NuSTAR* images of the Eel pulsar wind nebula, identifying synchrotron burnoff and evidence of a shock feature; results published in *ApJ*.
- Performed spectral and timing analysis of *NuSTAR* observations of HESS J0632+057; results published in *ApJ*.

Detector development for the GAPS experiment Aug. 2019–Feb. 2020
PI: Charles Hailey

- Fabricated, tested, and assembled Si(Li) detector-array modules for GAPS, an experiment designed to search for cosmic antimatter signatures of dark matter annihilation.

Undergraduate Research

ALS pathology in mouse and cell-culture models Sep. 2013–May 2014
Research Assistant, Columbia University Medical Center
PI: Christopher Henderson; Supervisor: Dima Yudin

- Used immunohistochemistry to measure muscle denervation over time and optimized embryonic stem-cell-derived motor-neuron cultures to test trophic factors for a senior capstone project.

Lung perfusion bioreactor Jan. 2011–May 2011
Undergraduate Researcher, The Cooper Union and Columbia University Medical Center
PIs: Eric Lima and Gordana Vunjak-Novakovic

- Designed and built a prototype lung perfusion bioreactor for testing on swine lung tissue.

UNIVERSITY TEACHING AND MENTORING

Yale University

Course Development Summer/Fall 2024
Introduction to Cosmology (ASTR 1700); Instructor: Priyamvada Natarajan

- Co-developed and redesigned an undergraduate cosmology course; wrote lecture slides, problem sets, and other materials that remain in use.

Graduate Teaching Fellow 2021–2026

- Instructed laboratories, held office hours, graded assignments, wrote solutions, and led individual review sessions.

Graduate Cosmology (PHYS 6000)	Spring 2026
Introduction to Mathematical Methods of Physics (PHYS 301, PHYS 4000)	Fall 2022, Fall 2025
Graduate Statistical Physics I (PHYS 512)	Spring 2024, Spring 2025
Graduate Classical Mechanics (PHYS 500)	Fall 2024
University Physics for the Life Sciences (PHYS 171)	Spring 2023
General Physics Laboratory II (PHYS 166L/S166)	Spring 2022, Summer 2022
General Physics Laboratory I (PHYS 165L)	Fall 2021

Graduate Student Mentor 2025–Present
Yale Undergraduate Mentorship Initiative

- Advise physics undergraduates on course selection, communication with faculty, and research opportunities through regular individual meetings.

Columbia University

Undergraduate Research Mentor 2020

- Mentored three undergraduate researchers on X-ray astronomy projects, providing training in data reduction, analysis, and academic writing.

The Cooper Union

Teaching Assistant Fall 2011
Introductory Physics Laboratory (PH291)

SECONDARY-SCHOOL TEACHING

SAR High School

Full-Time Chemistry Teacher Spring 2021

- Taught four 10th-grade chemistry sections spanning remedial to honors levels.

Full-Time Physics Teacher and Advisor Fall 2014–Spring 2019

- Designed and taught differentiated physics and engineering curricula for students in grades 10–12, including four physics courses across two tracks for grades 11–12.
- Served as an academic advisor for 10th-grade students and taught Jewish philosophy and text study in grades 9 and 11.
- Developed and directed engineering electives and co-curricular programs; coached the robotics team to second place in the Technion's international Robotraffic competition and received its Teacher Award.
- Published five articles on pedagogy and science education for the school community blog.

Naaleh High School for Girls

STEM Teacher Fall 2019–Spring 2020

- Wrote and taught a 10th-grade course in computer programming and engineering design.
- Designed and led a Python coding co-curricular program affiliated with Girls Who Code.

ACADEMIC PRESENTATIONS

Invited Talks

1. "Core-collapsed SIDM halos as massive SMBH seeds," Hebrew University Cosmolunch; Apr. 24, 2025. Jerusalem, Israel.
2. "The Unified AGN Model in X-ray observations," Columbia University High-Energy Astrophysics Meeting; Jan. 28, 2022. Virtual.

Conference Talks

1. "Gravothermal Evolution with Multiple Mass Species," Modelling and Observing Dense Stellar Systems (MODEST26); Aug. 16–22, 2026; Quy Nhon, Vietnam.
2. "Core-collapsed SIDM halos as massive SMBH seeds," YCIU Workshop; May 30, 2025; New Haven, CT.
3. "Using BASS to Detect Obscuration Bias and Test AGN Fitting Models for Low Count Data," BASS2024: New Horizons for Understanding Nearby AGN; Apr. 22, 2024; Virtual.
4. "Quantifying AGN Selection Effects in the Chandra COSMOS-Legacy Survey with BASS," Accretion History of AGN (AHA) III Workshop; Dec. 15, 2023; Miami, FL.
5. "Quantifying Selection Effects in Deep-Field AGN Surveys," New England Regional Quasar and AGN Meeting (NERQUAM); May 6, 2022; Storrs, CT.
6. "A NICER X-ray View of White Dwarf Pulsar AR Scorpii," Spring 2021 NICER Data Analysis and Science Workshop; May 13, 2021. Virtual. Co-presented with N. Saffold. Video of presentation.

Conference Posters

1. **Tokayer, Y. M.**, Koss, M., Urry, C. M., et al., Quantifying Selection Effects in Deep-Field AGN Surveys with BASS. In: *241st Meeting of the AAS*, January 8–12, 2023. Seattle, WA.
   Link to poster.
2. Woo, J., An, H., Burgess, D., et al. (incl. **Tokayer, Y. M.**), Multi-wavelength Study of PeVatron Candidate Pulsar Wind Nebulae. In: *AAS/High Energy Astrophysics Division 19th Annual Meeting*, March 13–17, 2022. Pittsburgh, PA.
3. Saffold, N., **Tokayer, Y. M.**, Mori, K., A NICER X-ray View of White Dwarf Pulsar AR Scorpii. In: *237th Meeting of the AAS*, January 10–15, 2021. Virtual.
   Link to poster.

ACCEPTED OBSERVING PROPOSALS

1. Co-Investigator. Ref #GN-2024A-Q-138 for Gemini GMOS Observations. *Resolving Mrk 248: A Potential Triple AGN with a Hidden 500 pc Dual AGN*.
   PI: Michael Koss.
2. Co-Investigator. Proposal 8087 for NuSTAR Observations. *A Survey of the Most Luminous Hard X-ray Selected Obscured Quasars at z=0.2–0.4*.
   PI: Michael Koss.
3. Co-Investigator. Proposals 90296 and 094349 for XMM-Newton Observations. *A Survey of the Most Luminous Hard X-ray Selected Obscured Quasars at z=0.2–0.4*.
   PI: Michael Koss.

OUTREACH AND PUBLIC COMMUNICATION

Planetarium Presenter Jan. 2024–Present
Leitner Observatory and Planetarium, Yale University

- Write and present planetarium shows using night-sky simulations for audiences from pre-K students to adults, primarily serving local schools and organizations.
- Design and deliver astronomy lessons for K–12 classrooms and lead evening telescope demonstrations.

Astronomy on Tap Fall 2022
New Haven, CT

- Helped organize and run public astronomy nights featuring trivia and talks by members of the Yale Astronomy Department.

Super Science Showdown Spring 2022
Yale Open Labs

- Ran interactive science events for students in grades 6–8 in New Haven County.

Engineers as Teachers Spring 2011
Iridescent and The Cooper Union

- Developed interactive lessons on sound and music and taught them at family science nights at a New York City middle school.

Public Talks

1. "Are we alone? The new search for other worlds and habitable planets beyond the Solar System," SAR High School; Apr. 17, 2024; Riverdale, NY. Video of talk.
2. "The hype is real: what we are already learning from the most powerful telescope in human history," SAR High School; Jan. 4, 2023; Riverdale, NY. Video of talk.

ACADEMIC SERVICE AND COMMUNITY LEADERSHIP

Co-Founder and Organizer Sep. 2022–May 2026
History and Foundations of Physics Reading Group

- Co-founded and managed an interdisciplinary discussion group on the history and philosophy of science, with participants ranging from undergraduates to tenured faculty.
- Selected readings, recruited guest speakers, facilitated discussions, and managed communications and scheduling.
- Secured funding from multiple sources and managed the group's budget.

Board of Trustees 2023–2024
Slifka Center for Jewish Life at Yale

Graduate Student Representative Spring 2023
Physics Department Faculty Hiring Committee, Yale University

- Interviewed candidates and evaluated job talks for a tenure-track astrophysics position.
- Worked with three other representatives to recruit student participation and write a report summarizing student input.

Co-Founder and Co-Director 2019–2021
Machon Siach Graduate Fellowship

- Designed curriculum and led discussions for a year-long fellowship for young professionals; managed recruitment and admissions and recruited guest speakers.

Head of Staff 2018–2020
Camp Moshava

- Supervised and coordinated more than 300 staff across 30 departments in a residential summer camp serving approximately 900 campers.
- Led staff training and professional development.

CONFERENCES, WORKSHOPS, AND SCHOOLS ATTENDED

- **Modelling and Observing Dense Stellar Systems (MODEST26)**. August 16–22, 2026. Quy Nhon, Vietnam.
- **AI Winter School**. Brown University. January 6–9, 2026. Virtual.
- **Yale Center for the Invisible Universe (YCIU) Workshop**. Yale University. May 30, 2025. New Haven, CT.
- **Accretion History of AGN (AHA) III Workshop**. University of Miami. December 14–17, 2023. Miami, FL.
- **Cosmology Summer School**. University of Michigan. June 5–9, 2023. Ann Arbor, MI.
- **241st Meeting of the AAS**. January 8–12, 2023. Seattle, WA.
- **New England Regional Quasar and AGN Meeting (NERQUAM)**. University of Connecticut. May 6, 2022. Storrs, CT.
- **NICER Data Analysis and Science Workshop**. May 13, 2021. Virtual.

TECHNICAL SKILLS AND LANGUAGES

- **Programming:** Python, MATLAB, C++, Fortran, HTML/CSS.
- **Scientific software:** NASA HEASoft, Mathematica, LaTeX, LabVIEW, Excel.
- ***N*-body and dynamical modeling:** `gyrfalcON`, `Agama`.
- **Python libraries:** Astropy, Stingray, PyXSPEC, pandas, NumPy, Matplotlib.
- **Space-based telescope data:** *Chandra*, *Swift*, *NuSTAR*, *NICER*.
- **Observing experience:** Palomar (Double Spectrograph, TripleSpec); Keck (NIRC2+AO, OSIRIS+AO).
- **Engineering and design:** Arduino, Microchip PIC, AutoCAD, SolidWorks, laser cutting.
- **Languages:** English (native), Hebrew (fluent), Yiddish (basic), German (basic).